OPERATION

The brake pedal is attached to the booster push rod. When the pedal is depressed, the primary booster push rod is depressed which moves the booster secondary rod. The booster secondary rod depresses the master cylinder piston.

REMOVAL

(1) Remove negative battery cable.
(2) Remove brake lamp switch.
(3) Remove ABS controller if equipped.
(4) Remove retainer clip securing booster push rod to pedal (Fig. 46) and clutch rod retainer clip if equipped.
(5) Remove bolts from brake pedal support and booster mounting nuts. Remove mounting stud plate nuts or clutch cylinder mounting nuts if equipped.
(6) Slid brake booster/master cylinder assembly forward.
(7) Remove mounting stud plate or slid clutch cylinder forward if equipped.
(8) Tilt the pedal support down to gain shaft clearance.
(9) Remove pedal shaft C–clip from passenger side of the shaft.
(10) Slide the pedal shaft toward the drivers side and remove the remaining C-clip.
(11) Slid the shaft out of the pedal bracket and remove the pedal.
(12) Remove pedal bushings if they are to be replaced.

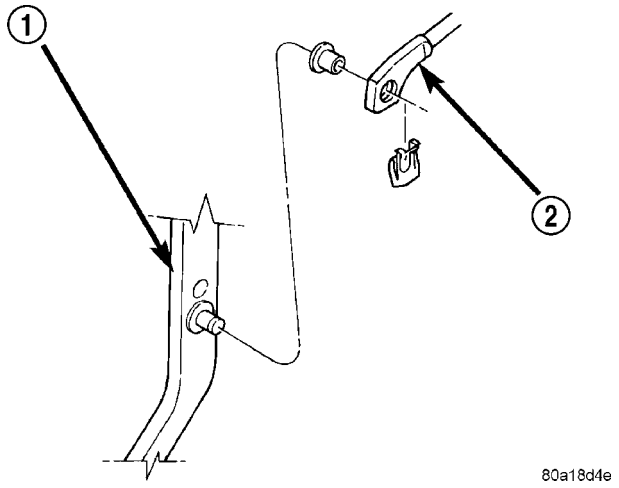

Fig. 46 Push Rod Attachment

1 - BRAKE PEDAL
2 - BOOSTER ROD

INSTALLATION

(1) Install new bushings in pedal. Lubricate bushings and shaft with multi-purpose grease.
(2) Position pedal in bracket and install shaft.
(3) Install new pivot pin C-clip.
(4) Position pedal support and install support bolts and tighten to 28 N·m (21 ft. lbs.).
(5) Slid the booster/master cylinder assembly into place, install mounting nuts and tighten to 39 N·m (29 ft. lbs.).
(6) Install stud plate or clutch cylinder if equipped and tighten mounting nut to 28 N·m (21 ft. lbs.).
Install retainer clip securing booster push rod to pedal (Fig. 46) and clutch rod retainer clip if equipped.
(7) Install ABS controller if equipped.
(8) Install and connect brake lamp switch.
(9) Install negative battery cable.

POWER BRAKE BOOSTER

DESCRIPTION

The booster assembly consists of a housing divided into separate chambers by two internal diaphragms. The outer edge of each diaphragm is attached to the booster housing.

Two push rods are used in the booster. The primary push rod connects the booster to the brake pedal. The secondary push rod connects the booster to the master cylinder to stroke the cylinder pistons.

OPERATION

The atmospheric inlet valve is opened and closed by the primary push rod. Booster vacuum supply is through a hose attached to an intake manifold fitting at one end and to the booster check valve at the other. The vacuum check valve in the booster housing is a one-way device that prevents vacuum leak back.

Power assist is generated by utilizing the pressure differential between normal atmospheric pressure and a vacuum. The vacuum needed for booster operation is taken directly from the engine intake manifold. The entry point for atmospheric pressure is through a filter and inlet valve at the rear of the housing (Fig. 47).

The chamber areas forward of the booster diaphragms are exposed to vacuum from the intake manifold. The chamber areas to the rear of the diaphragms, are exposed to normal atmospheric pressure of 101.3 kilopascals (14.7 pounds/square in.).

Brake pedal application causes the primary push rod to open the atmospheric inlet valve. This exposes the area behind the diaphragms to atmospheric pressure. The resulting pressure differential provides the extra apply force for power assist.

The booster check valve, check valve grommet and booster seals are serviceable.